COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 / Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	Aug 13 2026	DATE SUBMITTED:	Aug 13 2026
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ACTIVITY NO. Prelim Skills Exam

1. PROCEDURE OUTPUT

Hostname	Operating System	Ip Address
server1	Ubuntu	192.168.56.7
server2	Ubuntu	192.168.56.8
server3	Centos	192.168.56.9

Part A: Installing and Verifying Ansible

Submit a screenshot of the following:

- Ansible version

```
kenzo@workstation:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/kenzo/.ansible/plugins/modules', '/usr
/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
kenzo@workstation:~$
```

- **SSH connectivity**

```
kenzo@workstation:~$ ssh kenzo@server1
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Thu Jul 30 08:55:50 2026 from 192.168.56.6
kenzo@server1:~$
```

```
kenzo@workstation:~$ ssh kenzo@server2
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Last login: Thu Jul 30 08:56:24 2026 from 192.168.56.6
kenzo@server2:~$
```

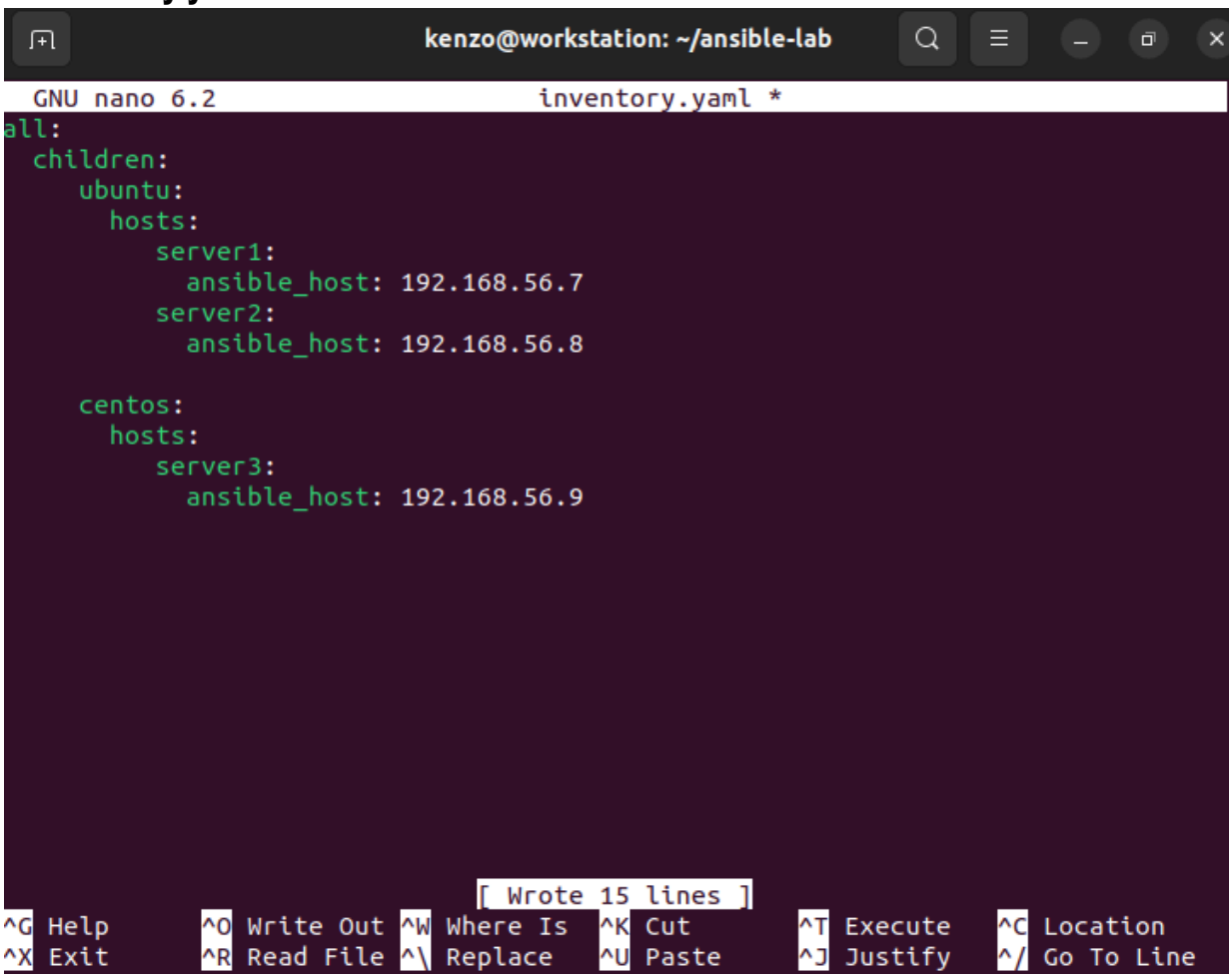
```
kenzo@workstation:~$ ssh kenzo@server3
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Aug 13 08:20:10 2026 from 192.168.56.6
[kenzo@server3 ~]$
```

Part B: Building an Inventory File

Demonstrate the following:

- inventory.yaml



- Successful inventory verification

```
kenzo@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
    "_meta": {
        "hostvars": {
            "server1": {
                "ansible_host": "192.168.56.7"
            },
            "server2": {
                "ansible_host": "192.168.56.8"
            },
            "server3": {
                "ansible_host": "192.168.56.9"
            }
        }
    },
    "all": {
        "children": [
            "centos",
            "ubuntu",
            "ungrouped"
        ]
    },
    "centos": {
        "hosts": [
            "server3"
        ]
    },
    "ubuntu": {
        "hosts": [
            "server1",
            "server2"
        ]
    }
}
kenzo@workstation:~/ansible-lab$ ansible all --list-hosts
  hosts (3):
    192.168.56.7
    192.168.56.8
    192.168.56.9
```

- **Successful ping results**

```
kenzo@workstation:~/ansible-lab$ ansible all -m ping
192.168.56.8 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
192.168.56.9 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
192.168.56.7 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C: Creating a Metagroup

Demonstrate successful execution of the metagroup.




```
kenzo@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.
yaml
  hosts (3):
    server1
    server2
    server3
kenzo@workstation:~/ansible-lab$
```

Part D: Creating Variables

Demonstrate successful use of variables.

```
kenzo@workstation: ~/ansible-lab
  GNU nano 6.2                          testvars.yaml
---

- name: Display Variables
  hosts: linuxservers
  tasks:
    - name: Show Operating System
      debug:
        msg: "Host {{ inventory_hostname }} is running {{ os_type }}"
```

```
TASK [Show Operating System] ***************************************************
ok: [server1] => {
    "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
    "msg": "Host server2 is running Ubuntu"
}
ok: [server3] => {
    "msg": "Host server3 is running CentOS"
}

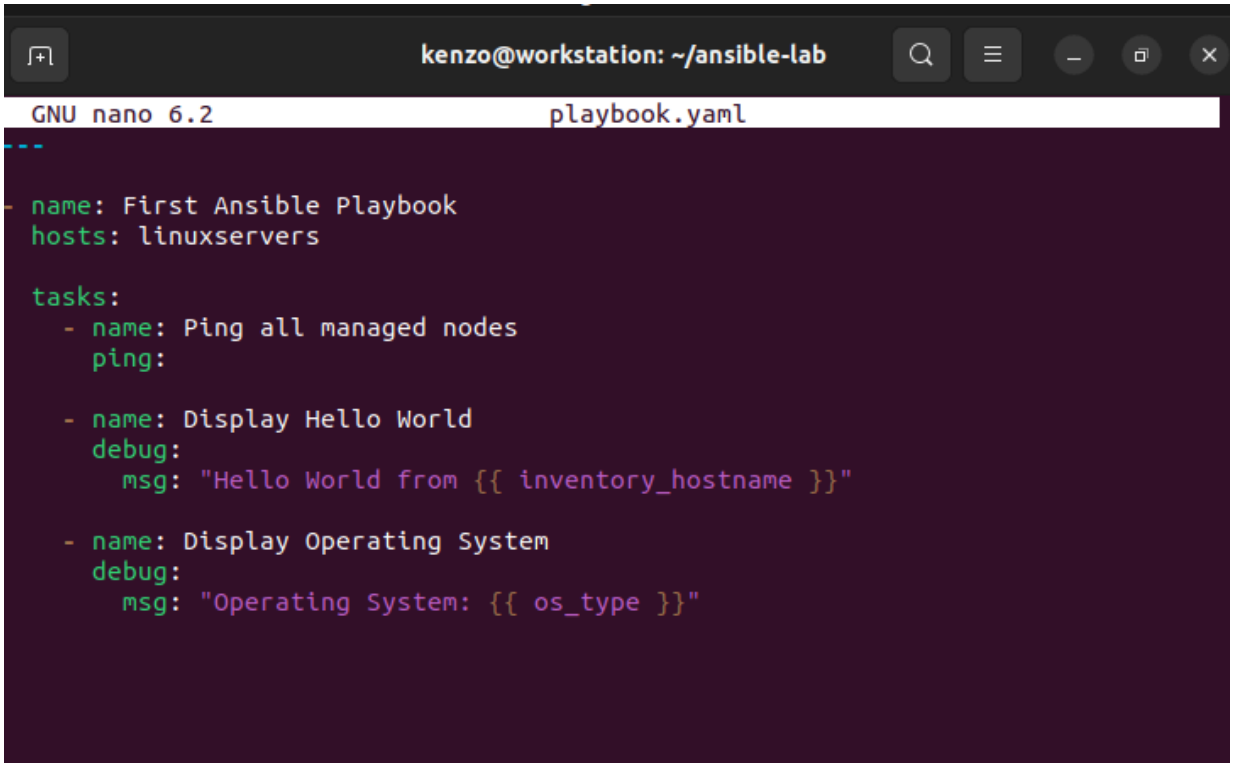
PLAY RECAP *********************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

kenzo@workstation:~/ansible-lab$
```

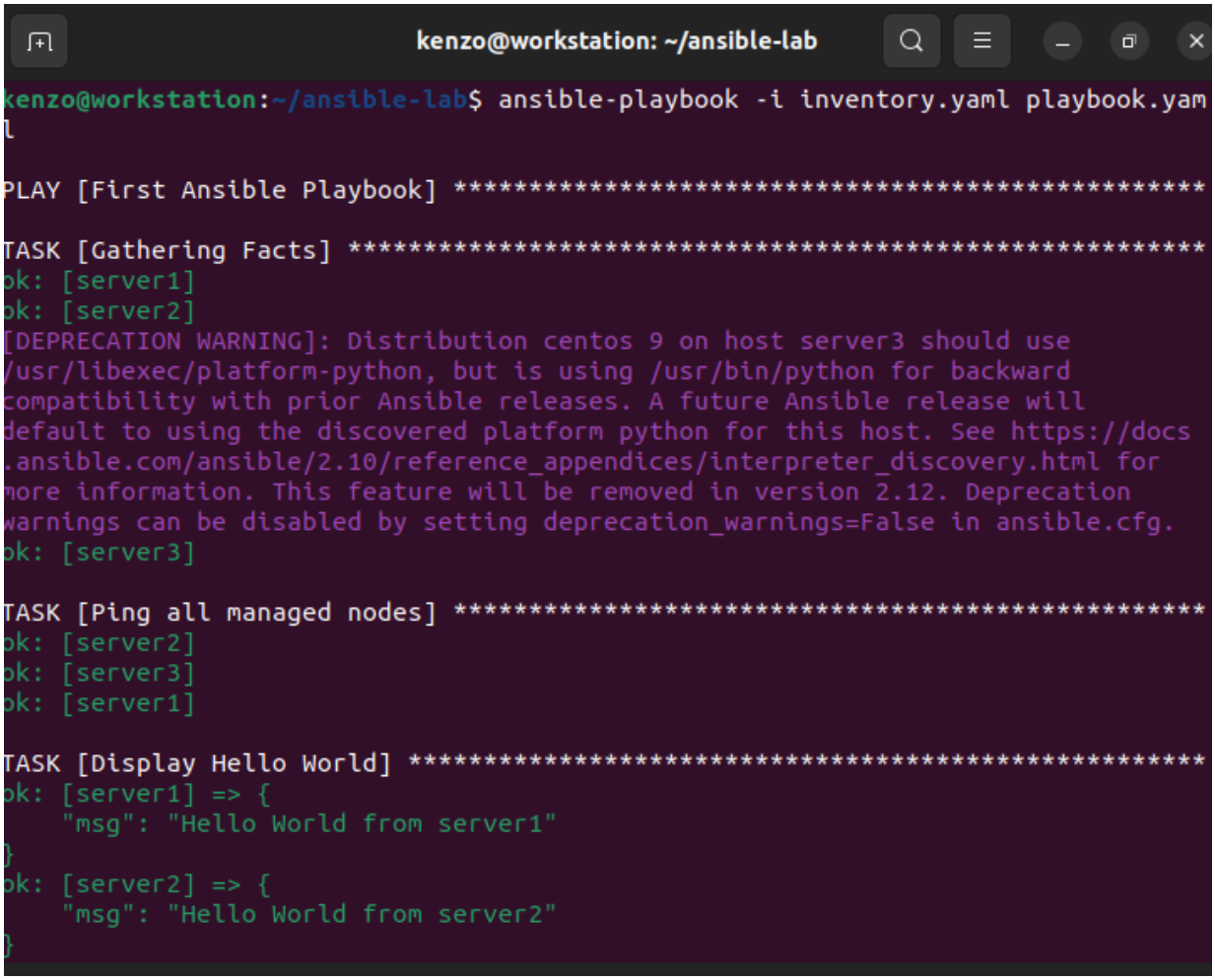
Part E: Creating Your First Playbook

Demonstrate:

- **Successful playbook execution**
- **Successful task completion on all managed nodes**


kenzo@workstation: ~/ansible-lab
GNU nano 6.2
playbook.yaml

- name: First Ansible Playbook
hosts: linuxservers
tasks:
- name: Ping all managed nodes
ping:
- name: Display Hello World
debug:
msg: "Hello World from {{ inventory_hostname }}"
- name: Display Operating System
debug:
msg: "Operating System: {{ os_type }}"


kenzo@workstation: ~/ansible-lab
kenzo@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml
PLAY [First Ansible Playbook] ***
TASK [Gathering Facts] **
ok: [server1]
ok: [server2]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using /usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [server3]
TASK [Ping all managed nodes] ***
ok: [server2]
ok: [server3]
ok: [server1]
TASK [Display Hello World] ***
ok: [server1] => {
"msg": "Hello World from server1"
}
ok: [server2] => {
"msg": "Hello World from server2"
}

```
ok: [server3] => {
    "msg": "Hello World from server3"
}

TASK [Display Operating System] ********************************************
ok: [server1] => {
    "msg": "Operating System: Ubuntu"
}
ok: [server2] => {
    "msg": "Operating System: Ubuntu"
}
ok: [server3] => {
    "msg": "Operating System: CentOS"
}

PLAY RECAP *********************************************************************
server1                    : ok=4    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server2                    : ok=4    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server3                    : ok=4    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

kenzo@workstation:~/ansible-lab$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named servers that contains only:

- **ubuntu1**
- **centos1**

```
servers:
  hosts:
     server1:
     server3:
```

Create a playbook named challenge.yaml that displays the message:
Welcome to Ansible Laboratory
Run the playbook and demonstrate that only hosts in the servers group receive the message.

kenzo@workstation: ~/ansible-lab

```
 GNU nano 6.2                        challenge.yaml
---

  name: Supplementary Activity
  hosts: servers

  tasks:
   - name: Display Text
     debug:
       msg: "Welcome to Ansible Playbook"
```

```
kenzo@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.ya
ml

PLAY [Supplementary Activity] **************************************************

TASK [Gathering Facts] *********************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [Display Text] ************************************************************
ok: [server1] => {
    "msg": "Welcome to Ansible Playbook"
}
ok: [server3] => {
    "msg": "Welcome to Ansible Playbook"
}

PLAY RECAP *********************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    s
```

```
PLAY RECAP *********************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
server3                    : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
```

Guide Questions

1. What is the role of the Control Node in Ansible?

It is the central machine where Ansible is installed and It is used to manage and configure target managed nodes.

2. Why is an inventory file required?

Because It lists and groups the target hosts and IP addresses that ansible will communicate with, without it there would be no communication at all to automate.

3. What is the purpose of a metagroup?

A metagroup allows you to next multiple groups in a single group name, which simplifies variable management.

4. How do variables simplify Ansible administration?

Variables are able to simplify administration by replacing hardcoded values with reusable data, just as I did in the supplementary activity wherein servers group called the previous group since it was already declared.

5. What is the difference between an ad hoc command and a playbook?

Ad hoc commands execute a single task on target host without saving it, while a playbook is a structured YAML file that contains reusable automated tasks.

Reflection:

To conclude, I was able to set up Ansible and accomplish activities such as building an inventory file and creating metagroups, as well as creating variables. One important thing that I realized in this environment is that we must be cautious in the spacing or indentations of the syntaxes inside the files as it is sensitive. Overall this preliminary exam was a great introduction to Ansible and was able to improve my confidence and understanding regarding inventory files and playbooks.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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